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財務金融碩士學程

碩士論文計畫書  
MASTER'S THESIS PROPOSAL

隱形帳簿：量化東協未計量的平台經濟  
The Invisible Ledger: Quantifying ASEAN's Unmeasured Platform  
Economy

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# 1. Research Background and Motivation

Southeast Asia's digital economy grew from \$98 billion in 2020 to \$263 billion in 2024 (Google, Temasek and Bain 2024). Much of this activity is coordinated by digital platforms that record transactions, calculate payouts, and recognize only a portion of gross transaction value as their own revenue. The resulting distinction is economically ordinary from an accounting perspective, but potentially important from a tax-administration perspective: participant-facing receipts may be digitally recorded by a formal intermediary without appearing in the platform's own revenue base or in a routine participant-level third-party reporting channel.

This proposal calls that administrative information gap the “invisible wedge.” The term invisible refers to the information channel available to tax administration, not to criminal concealment. If a platform processes 100 units of transaction value and recognizes 10 as platform revenue, the remaining 90 units are still recorded in the platform ecosystem. They may include merchant receipts, driver payouts, inventory costs, refunds, taxes already paid, and other pass-through amounts. The proposal therefore does not interpret the residual as profit, taxable income, unpaid tax, or GDP value-added.

The central research problem is measurement. Existing platform filings disclose gross merchandise or transaction value (GMV/GTV), platform revenue, and in some cases country or segment information, but these disclosures are rarely assembled to quantify the scale of participant-facing value relative to the revenue formally booked by the intermediary. The thesis develops a transparent measure of this gap and then asks whether it persists over time, how it should be interpreted for fiscal administration, and whether changes in the measure have short-window financial-market relevance.

The project is already at an advanced empirical stage. Preliminary analysis has been completed using Grab, GoTo, and Shopee disclosures; those results are summarized only as feasibility evidence in Section 6. The proposal examination is therefore intended to review the research question, measurement design, inferential boundaries, and remaining work before the final degree defense.

# 2. Research Questions and Objectives

The thesis is organized around four linked research questions. The first two establish the measurement object and its persistence. The third assesses policy relevance without imposing a causal interpretation on limited policy data. The fourth asks whether the same accounting information appears to be priced by investors around earnings announcements.

| Research question                                                                                                                        | Purpose                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| RQ1. How large is the gap between platform-mediated GTV/GMV and revenue booked by the platform in the Indonesia-focused sample?          | Construct the core measurement and quantify the scale of the administrative information gap.     |
| RQ2. Does the transaction-revenue gap persist within platforms over time?                                                                | Distinguish a structural feature of platform intermediation from a one-year accounting snapshot. |
| RQ3. What does Malaysia's 2024 low-value-goods expansion show descriptively about expanding digitally observable tax bases?              | Provide a policy benchmark while avoiding an unsupported causal claim.                           |
| RQ4. Are ecosystem-ratio levels or adjacent-quarter changes associated with short-window abnormal returns around earnings announcements? | Test whether the measure also carries short-horizon financial-market information.                |

The primary objective is not to estimate unpaid tax. It is to quantify a transaction flow that is already recorded by platforms but may remain outside routine participant-level information reporting, and to establish where that measure is—and is not—informative.

### **3. Literature Review and Research Gap**

#### **3.1 Platform economics and the location of value**

Multisided-platform theory explains why platform revenue is not a sufficient measure of the economic activity coordinated through the platform. Rochet and Tirole (2003), Caillaud and Jullien (2003), Parker and Van Alstyne (2005), and Armstrong (2006) show that platform pricing balances participation and cross-group network effects rather than behaving like a simple markup over cost. Hagiu and Wright (2015) further distinguish multisided platforms from vertically integrated firms by locating decision rights and entrepreneurial effort with affiliated participants. These models imply that a low retained share can coexist with a large transaction ecosystem.

#### **3.2 Informality, platform work, and measurement**

The informal-economy literature usually relies on household surveys, firm surveys, administrative records, or indirect macroeconomic methods (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019). Platform-mediated work occupies an unusual position because transactions can be priced, recorded, and settled through a formal intermediary while the participant remains self-employed, unregistered, below reporting thresholds, or outside an automatic information channel. The International Labour Organization (2021, 2023) likewise treats platform work and informality as overlapping but non-identical categories.

This distinction defines the thesis gap. The object of interest is not the shadow economy in the conventional sense. It is a digitally recorded participant-facing flow whose administrative treatment may still resemble self-reported income even though the intermediary already possesses the underlying transaction record.

#### **3.3 Third-party reporting as tax capacity**

Public-finance research consistently finds that third-party information and remittance structures shape compliance. Kleven et al. (2011) report an evasion rate of 0.3 percent for income subject to third-party reporting versus 37 percent for self-reported income in their Danish audit experiment. Pomeranz (2015) shows how the VAT paper trail creates self-enforcement, Naritomi (2019) demonstrates the role of consumer-held records, and Kleven, Kreiner and Saez (2016) model firms as fiscal intermediaries. Jensen (2022) links the expansion of formal organizations and reporting capacity to the development of modern tax systems.

Platform-specific studies reinforce the relevance of administrative records. Collins et al. (2019) and Garin et al. (2023) show how information returns affect measured gig-work prevalence in the United States, while Barrios, Hochberg and Yi (2022) and Denes, Lagaras and Tsoutsoura (2025) connect gig-platform participation to business formation and entrepreneurship. The common implication is that

platform-mediated activity leaves economically meaningful records even when the participant is not treated like a conventional employee or firm.

### **3.4 Digital taxation and reporting rules**

Corporate digital-tax reforms and participant-level platform reporting address different tax bases. OECD Pillar One concerns the allocation of taxing rights over large multinational enterprises (OECD 2020b). By contrast, the OECD Model Rules for Reporting by Platform Operators and the European Union DAC7 regime require platforms to collect and report seller or service-provider information (OECD 2019, 2020a; European Union 2021). These rules are the closest policy analogue to the mechanism studied here: using the platform's existing records to improve participant-level visibility without treating platform revenue as the participant tax base.

### **3.5 Research gap**

The literature establishes that platforms coordinate value beyond their booked revenue and that third-party reporting can materially increase administrative visibility. What is less developed is a transparent measurement connecting these two observations for large Southeast Asian platforms. This thesis contributes by constructing an auditable transaction-to-revenue measure from primary platform disclosures, explicitly documenting scope and derivation, and then testing persistence, descriptive policy relevance, and short-window investor response.

## **4. Conceptual Framework**

The core measure is the ecosystem ratio. Let GTV denote gross transaction value (or GMV where that is the platform's terminology), and let R denote platform revenue recognized from commissions, fees, and directly billed services. Unbooked GTV is  $GTV - R$ . The ecosystem ratio is:

$$E = (GTV - R) / R$$

A higher ratio means that a larger amount of participant-facing transaction value is processed relative to each unit of revenue booked by the platform. The ratio is unit-free, but cross-platform levels must be interpreted cautiously because revenue-recognition conventions and business models differ. The most defensible comparisons are therefore (i) within a clearly documented platform/scope and (ii) across platforms only after numerator and denominator are scope-matched.

The conceptual distinction is central to the thesis. The measure is intended to describe the scale of an administrative information problem; it is not a direct tax base. Participant costs, filing thresholds, legal status, taxes remitted through other channels, and deductible expenses determine actual tax liability and cannot be inferred from GTV alone.

## **5. Data and Research Design**

### **5.1 Platform selection and data provenance**

Grab, GoTo, and Shopee are selected because they are large ASEAN-origin consumer platforms that directly intermediate transactions and provide public financial disclosures sufficient to observe or

transparently derive transaction value and platform revenue. Grab and Sea Limited are publicly listed in the United States and provide SEC filings and investor-relations releases; GoTo provides annual and quarterly disclosures through the Indonesia Stock Exchange and company investor relations.

| Platform / input | 2023 construction                                                                                                                    | Provenance / scope                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Grab             | Indonesia revenue is directly disclosed; Indonesia GTV is derived using the company-wide disclosed take rate.                        | Country revenue is direct; GTV is derived. The relevant company measure is Mobility + Deliveries / On-Demand GMV. |
| GoTo             | Group GTV and net revenue are taken directly from the FY2023 annual report.                                                          | Both inputs are direct, but Group scope includes activity outside Indonesia.                                      |
| Shopee           | Indonesia GMV is estimated from regional disclosures and market-share information; revenue is derived using the disclosed take rate. | Indonesia-focused but partly derived; Sea consolidated revenue is not used as the denominator.                    |

The core estimate is therefore described as Indonesia-focused rather than perfectly country-isolated. This wording is intentional: the GoTo row uses Group-level disclosure because no country-level GTV split is publicly reported.

## 5.2 Core measurement and sensitivity

For each platform, the study computes unbooked GTV and the ecosystem ratio from fiscal-year 2023 values. A blended retained share is calculated as total platform revenue divided by total GTV across the three rows. The base blended take rate is 5.88 percent and is used only as a transparent calibration parameter for the four additional ASEAN markets. Sensitivity analysis varies the blended take rate from 5 to 10 percent while holding aggregate GTV fixed.

## 5.3 Historical persistence

The persistence analysis is deliberately separated from the Indonesia-focused cross-platform table. GoTo's 2020–2023 Group series is rebuilt from primary rupiah disclosures, with 2020–2021 based on the company's pro forma performance highlights. Grab historical observations use scope-matched Mobility + Deliveries GMV and revenue. Shopee historical observations pair e-commerce GMV with Shopee segment revenue. This design avoids mixing country-specific and company-wide series into one artificial panel.

## 5.4 Malaysia policy benchmark and regional calibration

Malaysia's January 2024 Low Value Goods (LVG) regime is used as a descriptive policy benchmark. The study reports the LVG regime's own observed collections and a two-country before-and-after comparison using Malaysia's SToDS + LVG collections and Vietnam's foreign-supplier portal collections. Because the comparison has only two countries, two years, and different tax bases, it is not treated as a difference-in-differences identification strategy.

For regional scale, the Indonesia-focused retained share is applied to 2024 digital-economy GMV in Vietnam, the Philippines, Thailand, and Malaysia. The resulting ASEAN-5 number is explicitly an order-of-magnitude calibration rather than five independently measured country wedges. All annual local-currency flows are converted using contemporaneous period-average official exchange rates from World Bank WDI / IMF International Financial Statistics.

## 5.5 Investor-response event study

The finance component asks whether ecosystem-ratio information is associated with short-window stock-market reactions when the underlying GMV/GTV and revenue figures are released. The event panel contains 47 firm-quarter observations from 2022Q1 through 2026Q1: 17 Grab, 16 GoTo, and 14 Sea observations, subject to disclosure availability. Event sessions are the first trading sessions able to react to each earnings release; after-close announcements are shifted to the next trading session.

Abnormal return is the platform's daily close-to-close return less its matched market-index return: the NASDAQ Composite for Grab, the Jakarta Composite Index for GoTo, and the S&P 500 for Sea. The primary dependent variable is  $CAR[-1,+1]$ . The ecosystem ratio is tested in levels and as adjacent-quarter percentage change. Because pooled level comparisons can reflect different reporting conventions, within-platform changes and post-structural-break specifications receive greater interpretive weight.

| Module            | Main variable / test                                           | Inferential role                                                        |
|-------------------|----------------------------------------------------------------|-------------------------------------------------------------------------|
| Core measurement  | $E = (GTV - Revenue) / Revenue$                                | Quantify the administrative information gap.                            |
| Persistence       | Scope-matched annual / historical series                       | Test whether the gap persists within platforms.                         |
| Policy benchmark  | Malaysia LVG + Malaysia/Vietnam descriptive comparison         | Illustrate administrative-base expansion without causal identification. |
| Investor response | Correlation of ecosystem ratio or QoQ change with $CAR[-1,+1]$ | Test short-window financial-market relevance.                           |

## 6. Preliminary Evidence and Feasibility

The following results are included to demonstrate feasibility and the current stage of the project. They are not presented as a substitute for the proposal committee's review of design and interpretation.

| Evidence already obtained   | Current result                                                                                                  | Interpretation                                                                                                |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 2023 Indonesia-focused base | \$71.0B GTV; \$4.18B platform revenue; \$66.83B unbooked GTV; ecosystem ratio 16.0×                             | The transaction-revenue wedge is large relative to booked platform revenue, but it is not a tax-due estimate. |
| GoTo 2020–2023 Group series | 97.6%–98.9% of GTV remains outside booked net revenue.                                                          | Supports persistence of a large gap rather than a monotonic increase.                                         |
| Malaysia 2024 benchmark     | RM476M LVG collections (~US\$104M) and ~US\$136M descriptive difference in changes vs Vietnam.                  | Policy benchmark only; tax bases and pre-trends are not matched.                                              |
| 47-event investor panel     | No robust relationship between ecosystem-ratio levels / QoQ changes and $CAR[-1,+1]$ in primary specifications. | Suggests administrative relevance does not automatically imply a short-horizon pricing signal.                |

A separate regional calibration applies the 5.88 percent retained share to four additional ASEAN digital-economy GMV figures and produces an ASEAN-5 order-of-magnitude wedge of approximately \$202 billion. Because the four non-Indonesian rows use economy-wide 2024 GMV while the Indonesia base is a 2023 three-platform construction and includes Group-level GoTo data, this figure is treated only as a transparent scenario.

## 7. Expected Contribution, Limitations, and Work Plan

### 7.1 Expected contribution

The expected contribution is primarily measurement and administrative interpretation. The thesis links platform-economics theory, which explains why booked revenue can be a thin share of ecosystem

activity, with the public-finance literature on third-party information reporting. It provides a reproducible measure of transaction value that is digitally recorded by an intermediary but not represented by the intermediary’s own revenue.

The policy implication is intentionally sequenced: establish reliable participant-level reporting first, distinguish participant types and thresholds second, and only then evaluate withholding or liability rules. This approach is consistent with the distinction between platform reporting regimes such as DAC7 and corporate digital-tax reforms that address different tax bases.

## 7.2 Main limitations

The core limitations are data scope and tax-base interpretation. GoTo’s 2023 input is Group-level; Grab and Shopee Indonesia inputs require transparent derivation; the regional calibration mixes 2023 platform data with 2024 economy-wide GMV; and GTV includes costs, pass-through amounts, and receipts that cannot be equated with profit or taxable income. The Malaysia comparison is descriptive, and the event panel is small and structurally heterogeneous. These limitations are part of the design rather than hidden assumptions.

## 7.3 Remaining work before the final degree defense

The empirical analysis is substantially complete. The proposal examination will be used to confirm the measurement boundaries, scope labels, and interpretation strategy. Remaining work consists of incorporating committee feedback, performing any additional robustness or provenance checks requested by the committee or advisor, finalizing the thesis presentation, and preparing the final manuscript for the degree oral examination.

| Period                                | Planned work                                                                                                   |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------|
| September 2026                        | Proposal oral examination; document committee comments; confirm final scope and robustness requests.           |
| October–November 2026                 | Complete requested checks; finalize tables, citations, and provenance; advisor review and manuscript revision. |
| December 2026 / advisor-approved date | Final degree oral examination and thesis submission, subject to program scheduling requirements.               |

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